

ARSHAM VASHISTH

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[LinkedIn](#) 

[Portfolio](#) 

DATA SCIENCE | ANALYTICS | AI | ML

MOTIVATION

Data Science and Analytics professional with a Master of Artificial Intelligence and hands-on experience applying Python, SQL, statistical analysis and machine learning to real-world problems. Skilled in analysing complex datasets, identifying patterns, testing hypotheses and translating analysis into practical insights. Curious, detail-oriented and motivated by solving problems where data can improve business outcomes.

SKILLS & TOOLS

Programming: Python, SQL, R, C, C++, Java

Data Science & Statistics: Statistical Analysis, Hypothesis Testing, A/B Testing, Predictive Modelling, Classification, Regression, Clustering, PCA, Causal Impact Analysis

Machine Learning: Scikit-learn, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Tools & Technologies: Git, Jupyter Notebook, Anaconda

Data & Visualisation: Tableau, Excel, Pandas, NumPy, Jupyter

Mathematics: Linear Algebra, Probability, Statistics, Experimental Design

PROJECTS

Enhancing Targeting Accuracy Using Machine Learning

- By analyzing historical customer behavior and campaign data, developed a [machine learning](#) model to predict the likelihood of customers signing up for a \$100 annual membership. This enabled the retailer to identify high conversion customers, reduce mailer distribution costs, and significantly [improve return on investment \(ROI\)](#). The solution provided insights for more targeted, cost-effective marketing strategies.

Predicting Customer Loyalty Using Machine Learning

- Led the development of a predictive model to estimate customer loyalty scores for a grocery retailer's untapped customer base. After market-level loyalty [data was only available for 50% of the customer database](#), I built a machine learning solution to identify key [relationships between customer behavior and loyalty metrics](#). This enabled the retailer to predict accurate loyalty scores for the remaining customers, ensuring [better customer segmentation, targeted marketing, and more effective communication](#) strategies ultimately improving customer retention and engagement.

Assessing Campaign Performance Using Chi-Square Test For Independence

- Conducted an [A/B test](#) for a grocery retailer's promotional campaign of their new "Delivery Club" membership program. Customers were randomly divided into three groups: low-cost mailer, high-cost mailer, and a control group. [Analyzed the differences in signup rates between the mailer types](#) to determine the most effective marketing strategy. The insights generated provided the client with clear recommendations for optimizing future campaigns, allowing for smarter marketing spend allocation and improved ROI.

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DATA SCIENCE | MACHINE LEARNING | AI

PROJECTS (CON'T)

Global Earthquake Tracking Dashboard (Tableau)

- Developed an interactive Tableau dashboard that visualizes [global earthquake activity over a 30 day period](#). The dashboard enables users to explore key patterns, magnitudes, and regional distributions of seismic events. By focusing on intuitive design and visually engaging features, it provides quick insights into earthquake [frequency, intensity, and geographical trends](#), empowering users to analyze and interpret seismic data with ease.

EXPERIENCE

Machine Learning Engineer, Intern - Silverpond

JUNE 2025 - DEC 2025

- Analysed and prepared [complex datasets](#) for machine learning applications, contributing to [model development, evaluation and improvement](#) across multiple use cases.
- Developed and evaluated [machine learning models](#) in Python, applying data preparation, [feature engineering and performance analysis](#) to improve model outcomes.
- Built a semantic segmentation solution for agricultural data, achieving an [IoU of 0.87](#) while improving [annotation efficiency by 30%](#).
- Investigated model performance and error patterns to identify opportunities for improvement, contributing to an overall [~18% improvement in model performance](#).
- Collaborated with cross-functional teams to develop and integrate machine learning solutions across diverse datasets and production pipelines.

EDUCATION

Master of Artificial Intelligence

2021 - 2023 - RMIT University

Bachelors of Computer Science and Engineering

2016 - 2020 - SRM University

REFERENCES

Jonathan Chang

Chief Technology Officer
Silverpond

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Joshua Patterson

Machine Learning Engineer
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